

# Polar coordinates and circular motion

# Key topics

Coordinate transforms

Trig identities

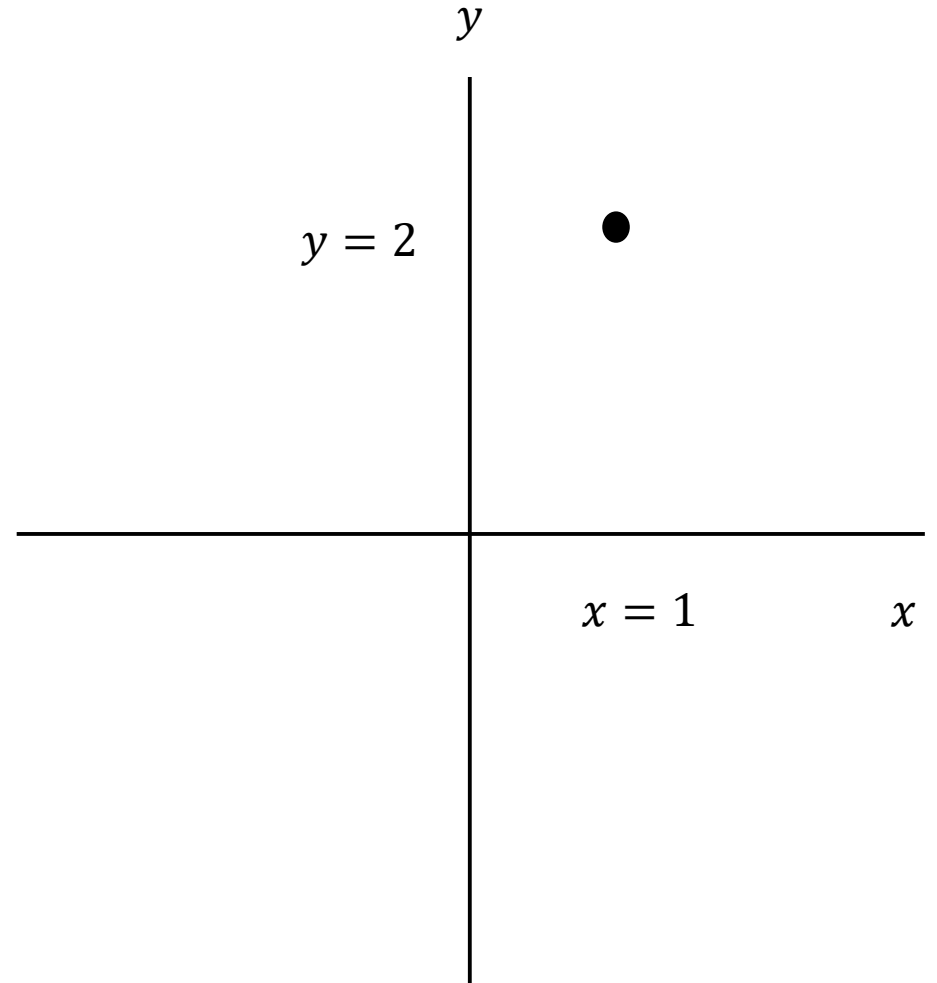
Polar curves

# Cartesian Coordinates

Cartesian coordinates are the first ones you learn, e.g. (1,2)

They are the basis of how we have defined graphing, vectors etc.

Without this system, we'd struggle to differentiate curves and define slopes

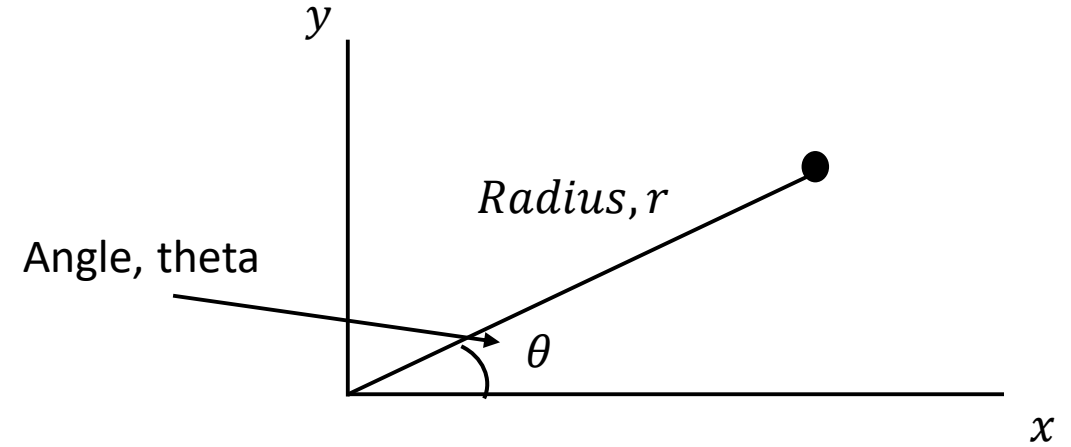


# Polar Coordinates

Polar coordinates are another way to define position, they have applications in complex numbers, waveforms etc.

They are the basis of how we interpret circular motion

Without this system, we'd struggle to navigate on a curved planet



# Conversion between cartesian and polar

The position can be defined in either way, so there must be a method for converting from one to the other

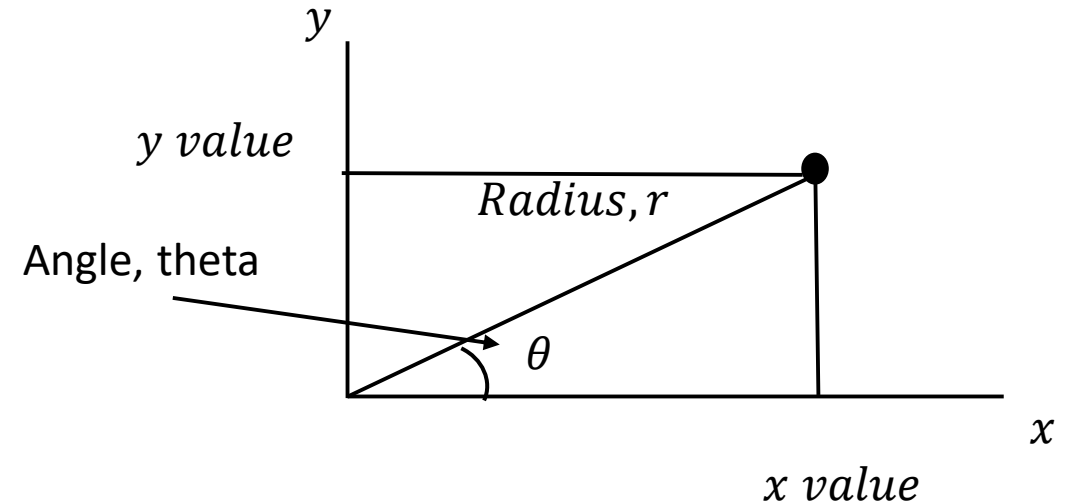
In fact, we can find this from our trig class:

$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{\textit{Opposite}}{\textit{Adjacent}}$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \tan^{-1} \frac{y}{x}$$



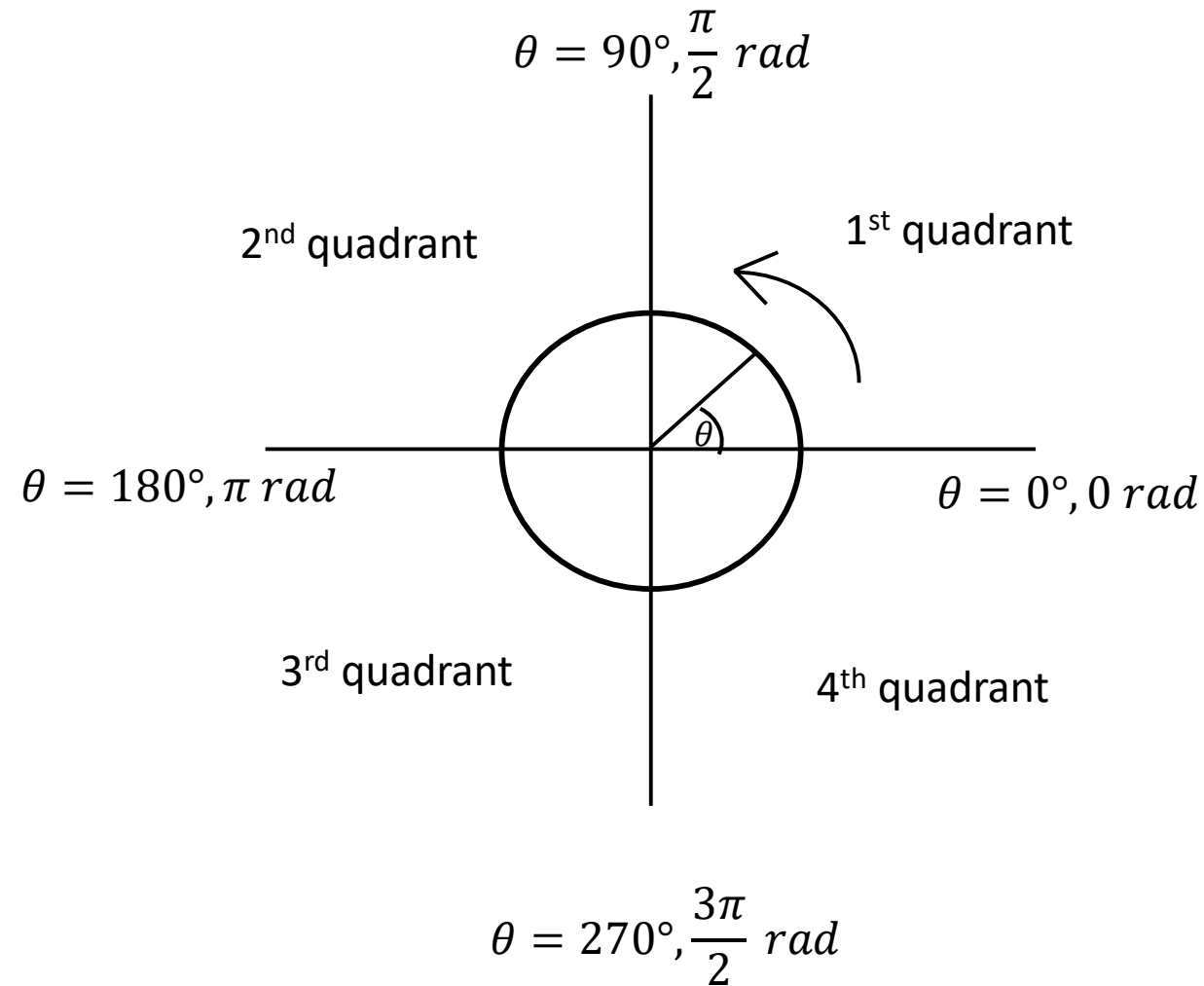
# Degrees Vs. Radians

Always use radians for calculations.

There are times when you'll calculate in radians but present your results in degrees (e.g. questions where the context involves Navigation)

I know it seems weird, but you use feet and inches for height, metres and centimetres for nearly everything else!

If I ask for one format of answer, use that one. If I don't specify, either is valid



# Finding theta

Consider this calculation:

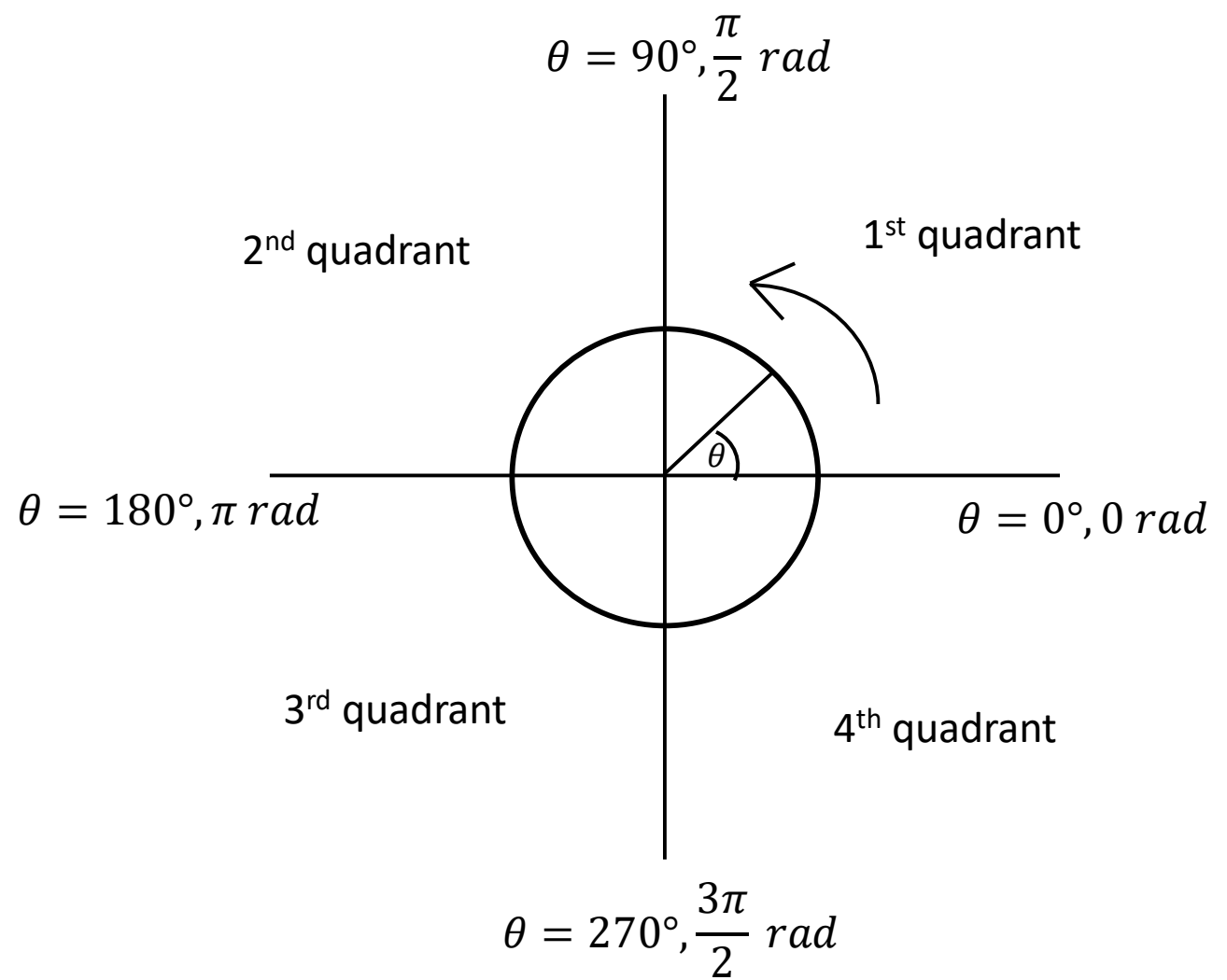
$$\theta = \tan^{-1} \frac{1}{1} = \tan^{-1} 1 = 45^\circ, \frac{\pi}{4} \text{ rad}$$

$$\theta = \tan^{-1} \frac{1}{-1} = \tan^{-1} -1 = 135^\circ, \frac{3\pi}{4} \text{ rad}$$

$$\theta = \tan^{-1} \frac{-1}{-1} = \tan^{-1} 1 = 225^\circ, \frac{4\pi}{3} \text{ rad}$$

$$\theta = \tan^{-1} \frac{-1}{1} = \tan^{-1} -1 = 315^\circ, \frac{7\pi}{4} \text{ rad}$$

Clearly, we are doing something wrong!



# Quadrants example

A=(1,2) in quadrant 1,

$$\text{so: } \theta_A = \tan^{-1} \frac{2}{1} = 63.43^\circ, 1.107 \text{ rad}$$

B=(-1,2) in quadrant 2,

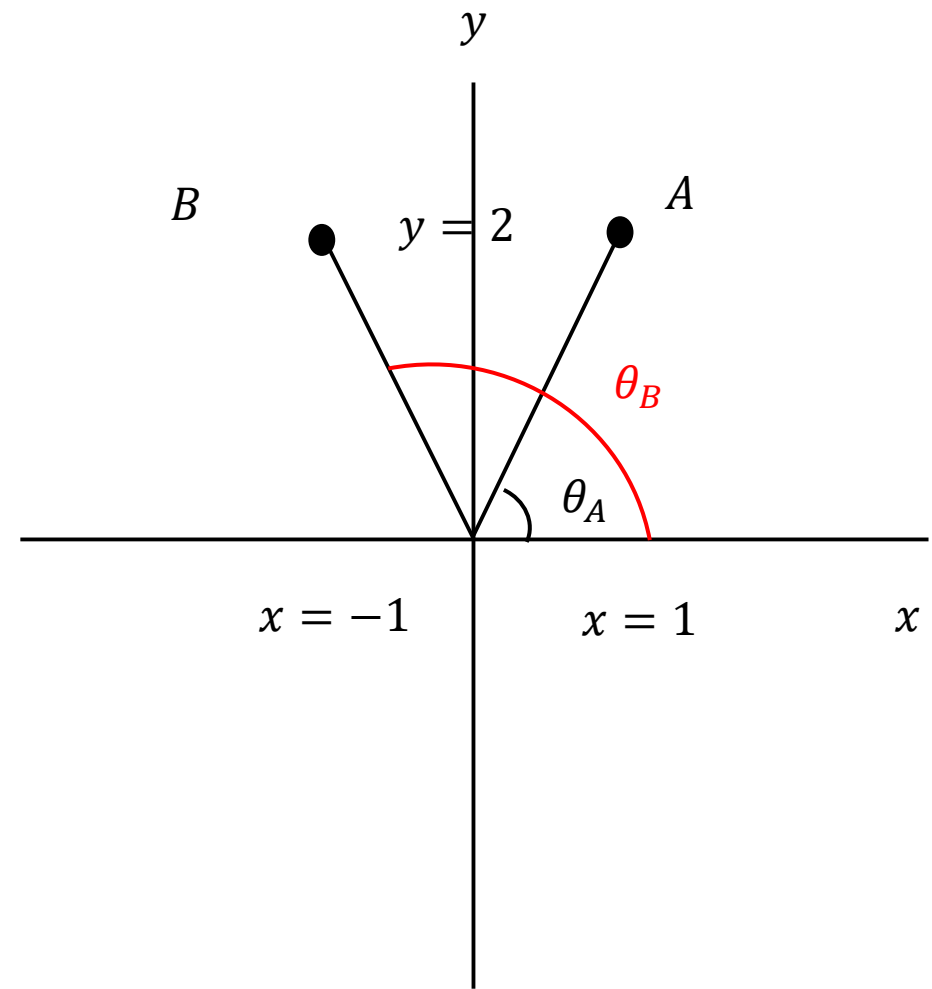
$$\text{so: } \theta_B = \tan^{-1} \frac{2}{-1} = -63.43^\circ, -1.107 \text{ rad}$$

**Wrong!**

B=(-1,2) in quadrant 2,

$$\text{so: } \theta_B = \tan^{-1} -\frac{2}{1} + 180^\circ \text{ or } \pi = -63.43^\circ + 180^\circ, -1.107 + \pi \text{ rad}$$

$$\theta_B = \tan^{-1} -\frac{2}{1} + 180^\circ \text{ or } \pi = 116.57^\circ, 2.03 \text{ rad}$$

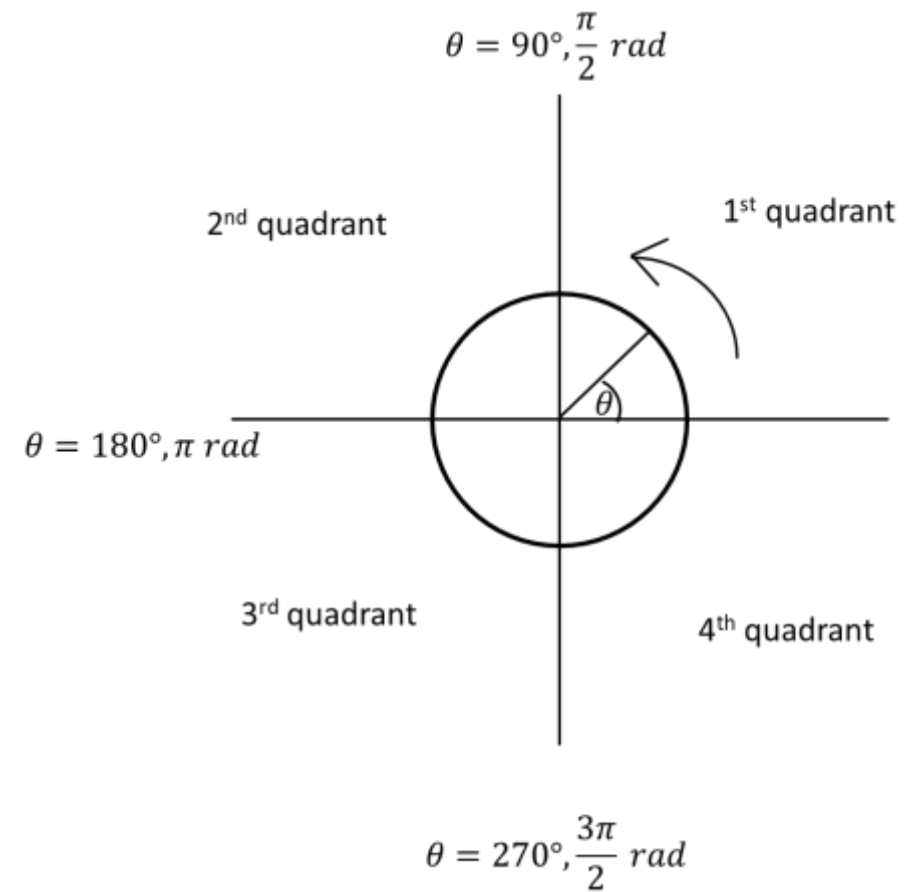


# Summary

We can translate between Cartesian and Polar using:

$$\theta = \tan^{-1} \frac{y}{x}, \quad x = r \cos \theta, \quad y = r \sin \theta$$

Adding  $\pi$  if we are in the second or third quadrant and  $2\pi$  if we are in the fourth



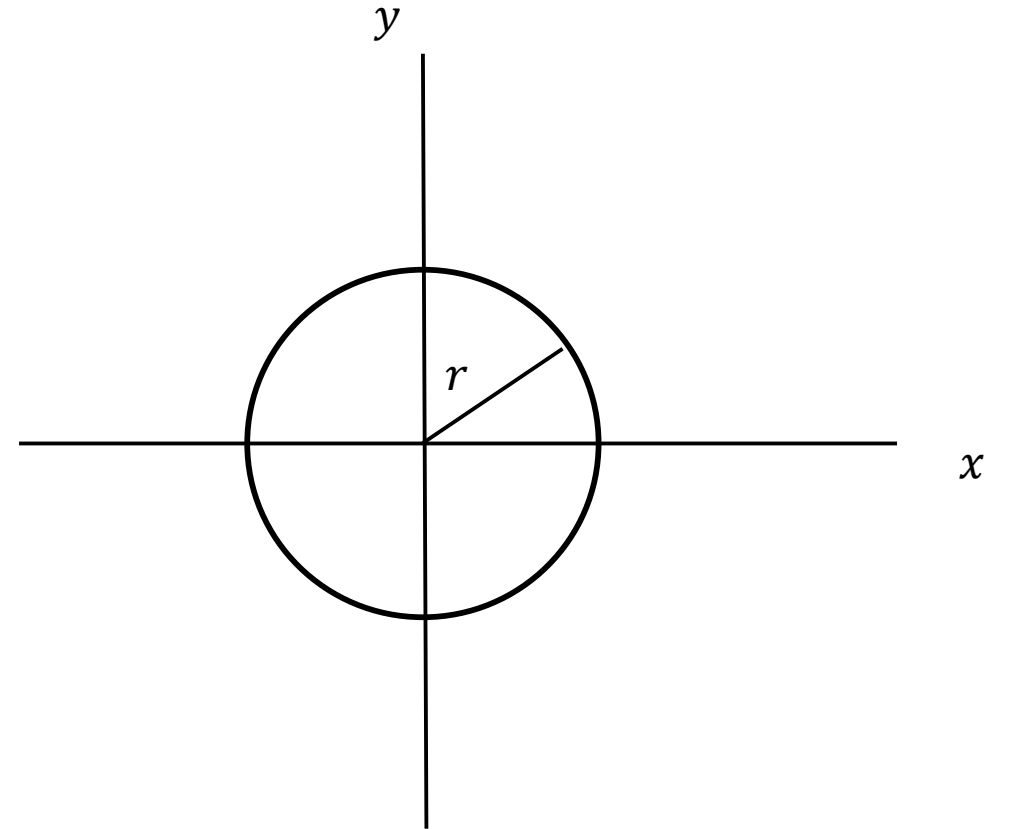
# Polar curves 1

We can graph functions in the Cartesian plane so we should be able to do the same in the Polar plane

Consider what happens if we keep a constant  $r$  and vary  $\theta$ :

We sweep out a circle of radius  $r$

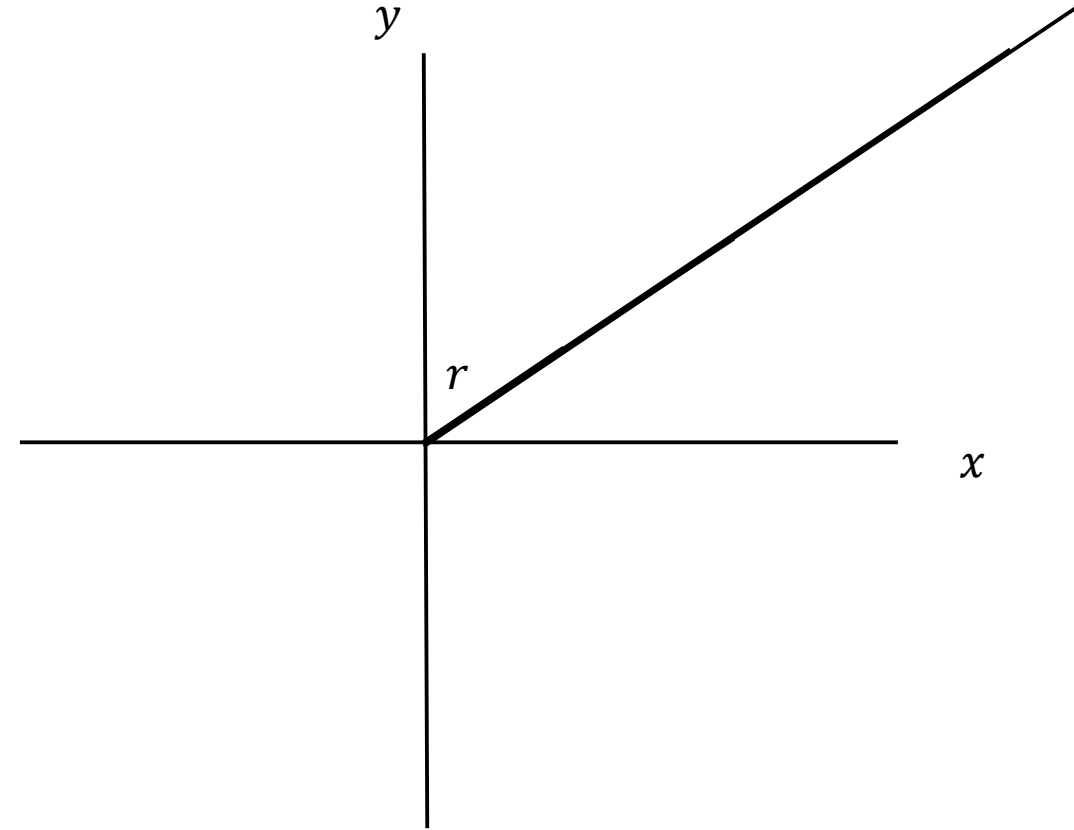
$$r = \sqrt{x^2 + y^2}$$



# Polar curves 2

Now imagine we keep theta constant and gradually increase the radius

We form a straight line (a ray) with intercept zero

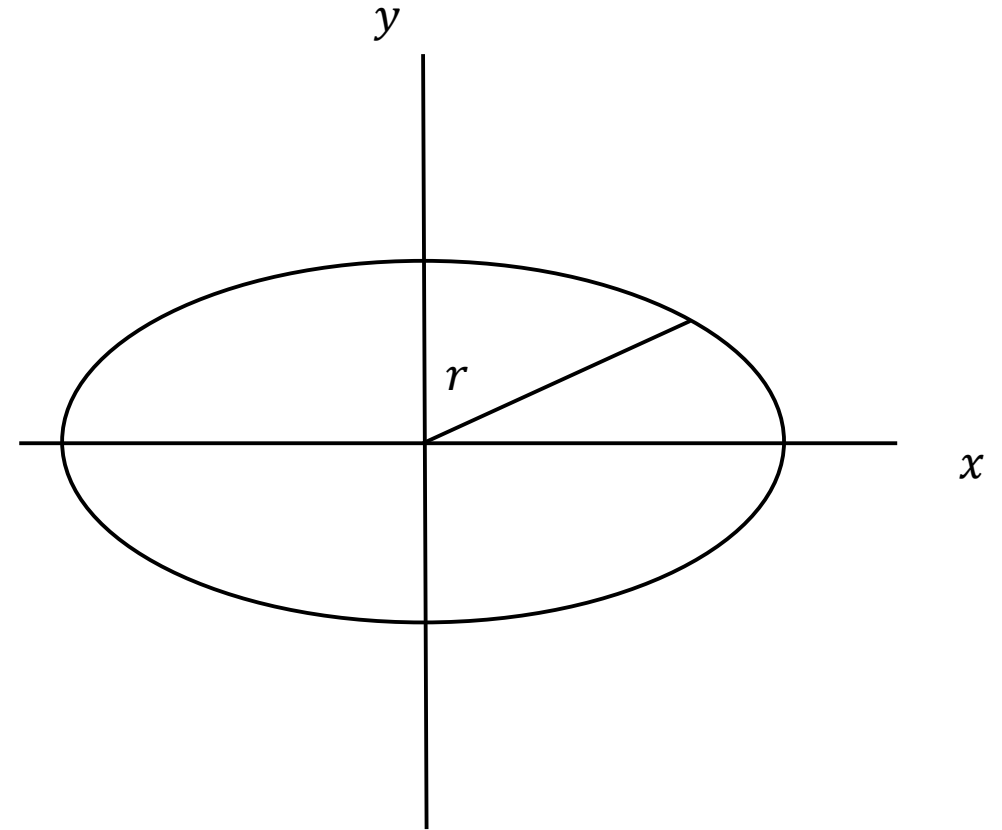


# Polar curves 3

Now consider changing both  $r$  and  $\theta$

Here the radius changes smoothly, as a function of the angle,  $\theta$

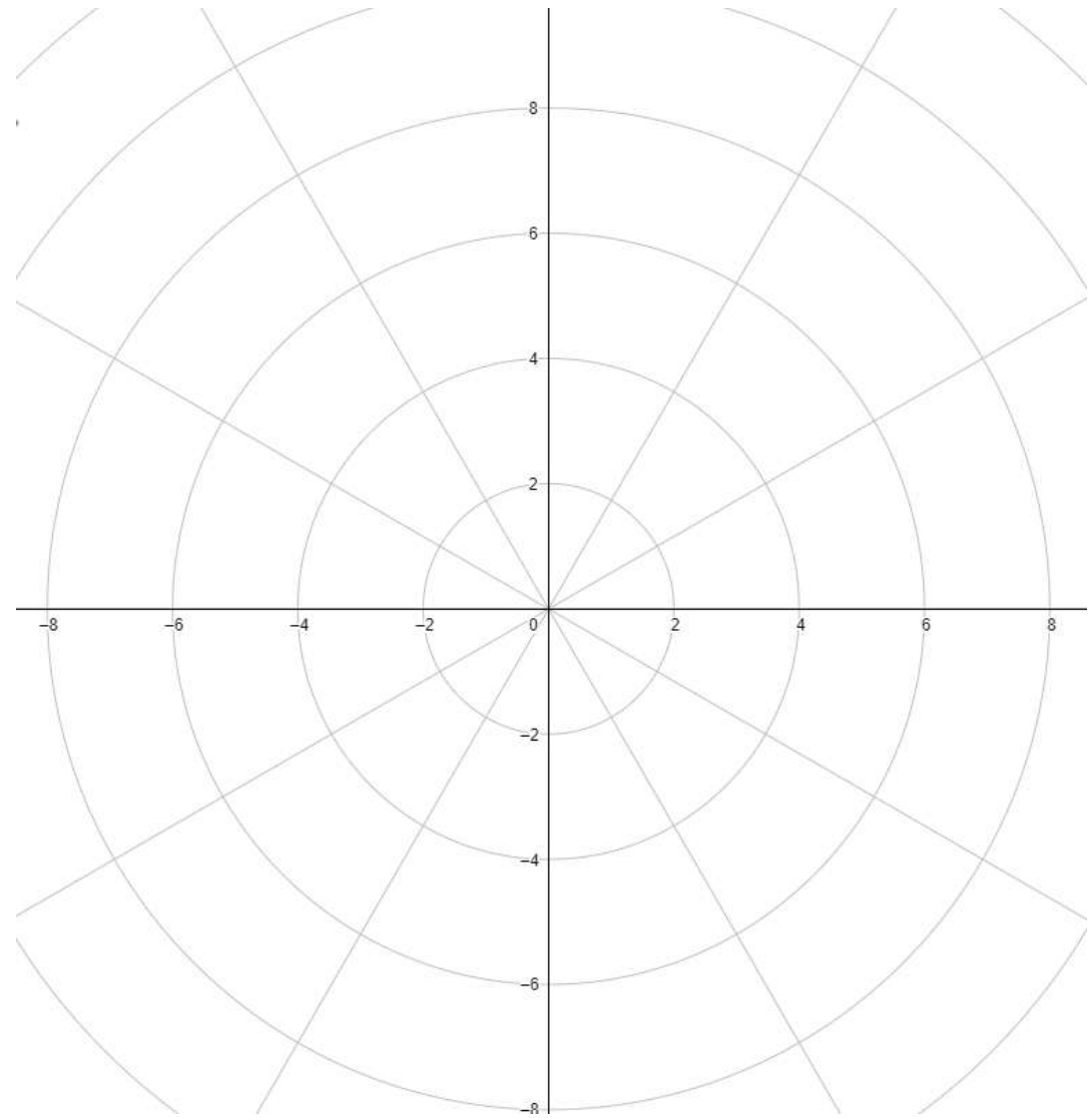
$|r|$  is large when  $\theta$  is close to zero and  $\pi$ , smaller when it is close to  $\frac{\pi}{2}$  and  $\frac{3\pi}{2}$



# Displaying Polar curves

We can change how we display our coordinate system to better find  $r$  and theta values.

The  $X$  and  $Y$  axes are still the same, but instead of highlighting the vertical and horizontal grid we highlight the angular and radial grid.



# Converting between Polar and Cartesian

$$x^2 + 2y^2 = 9$$

In fact, we can make use of trig to simplify this

$$x = r \cos \theta, y = r \sin \theta$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$(r \cos \theta)^2 + 2(r \sin \theta)^2 = 9$$

So we can rewrite:

$$r^2 = \frac{9}{\cos^2 \theta + 2 \sin^2 \theta}$$

$$r^2(\cos^2 \theta + 2 \sin^2 \theta) = 9$$

As:

$$r^2 = \frac{9}{\cos^2 \theta + 2 \sin^2 \theta}$$

$$r^2 = \frac{9}{1 + \sin^2 \theta}$$

$$r = \frac{3}{\sqrt{\cos^2 \theta + 2 \sin^2 \theta}}$$

$$r = \frac{3}{\sqrt{1 + \sin^2 \theta}}$$

# Extended trig relations list

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2 \cos^2 A - 1 \\ &= 1 - 2 \sin^2 A\end{aligned}$$

# Converting equations

The goal is to introduce the transforms and remove  $x$  and  $y$

$$y = 2x + 2$$

$$r \sin \theta = 2r \cos \theta + 2$$

$$r \sin \theta - 2r \cos \theta = 2$$

$$r (\sin \theta - 2 \cos \theta) = 2$$

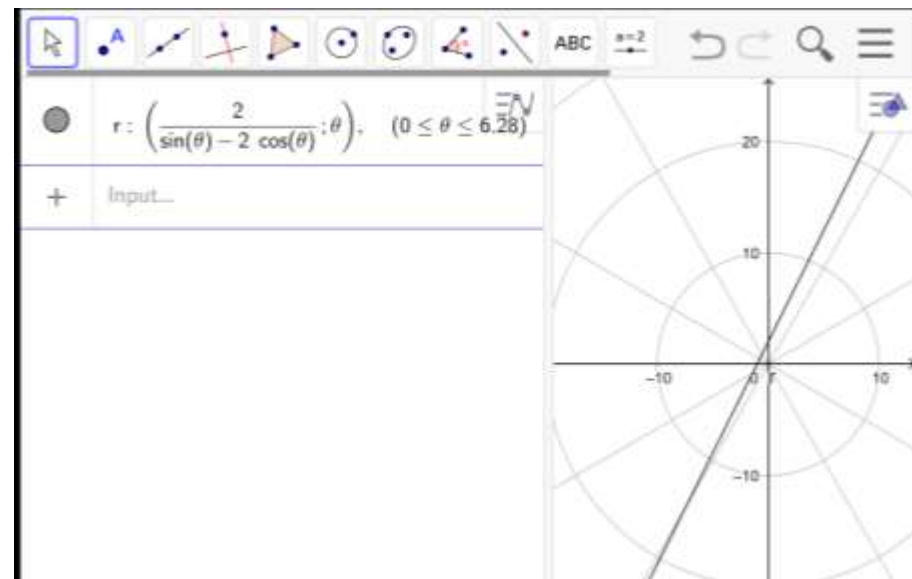
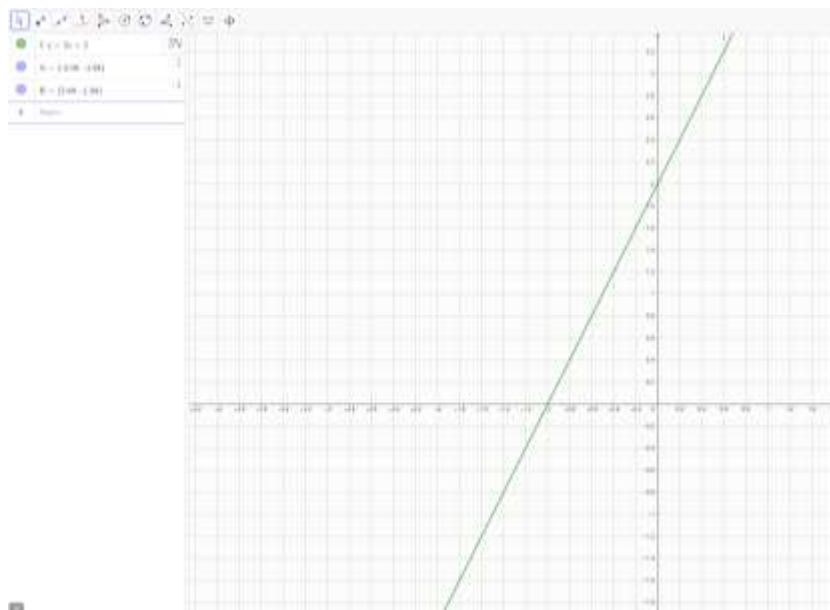
$$r = \frac{2}{(\sin \theta - 2 \cos \theta)}$$

Then collect terms to isolate  $r$

Then rearrange the equation to make  $r$  the subject

$$y = 2x + 2,$$

$$r = \frac{2}{\sin \theta - 2 \cos \theta}$$

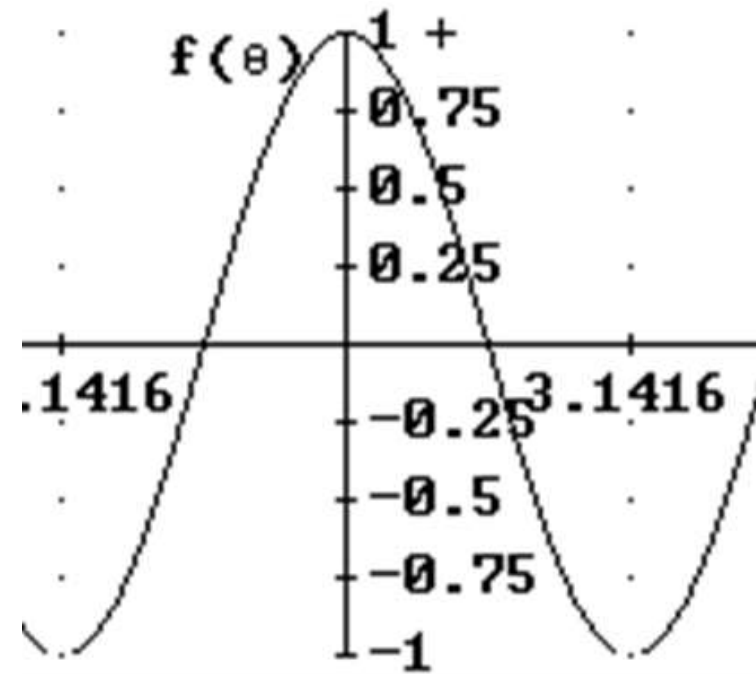


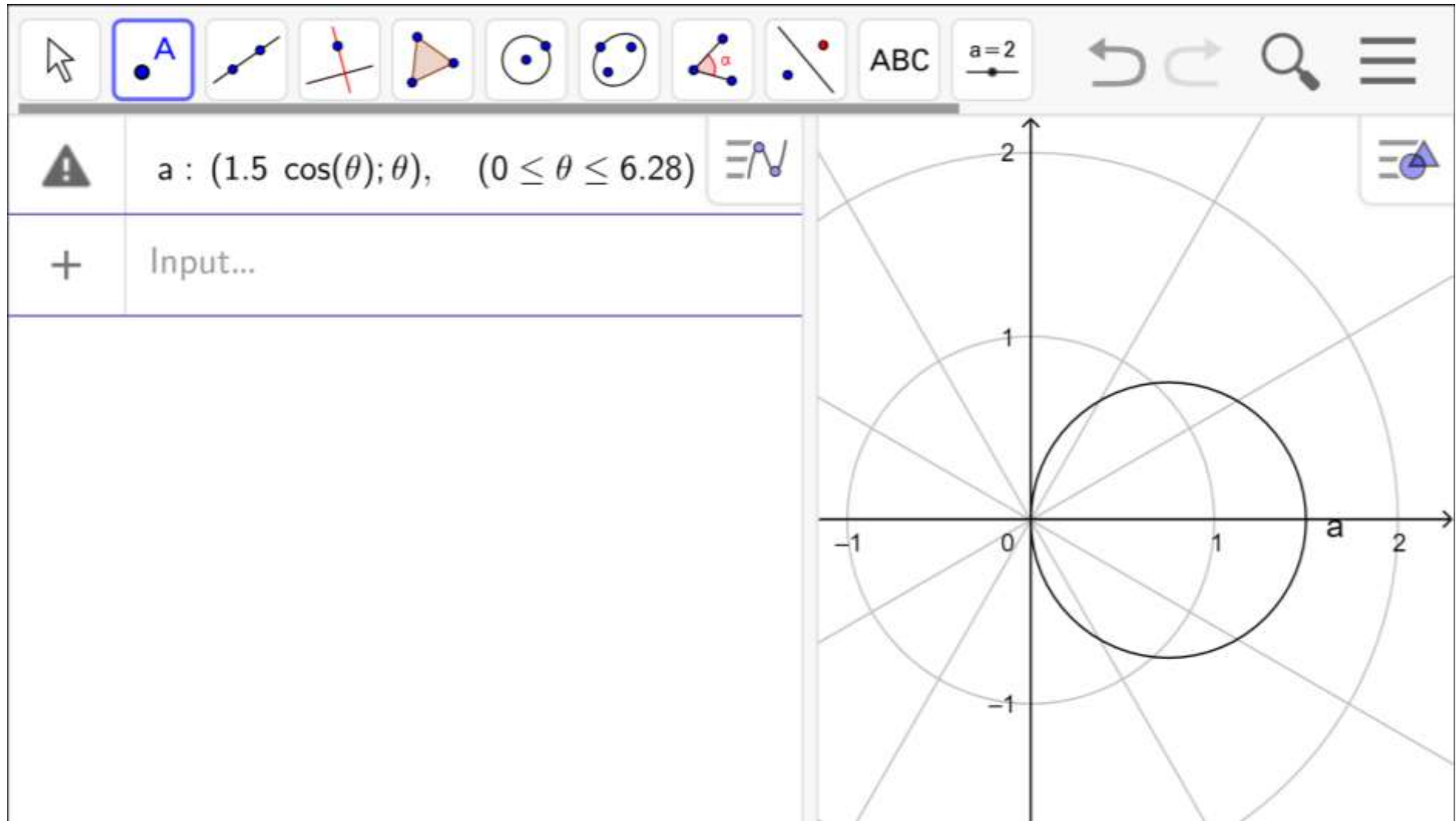
# Advanced Polar curves

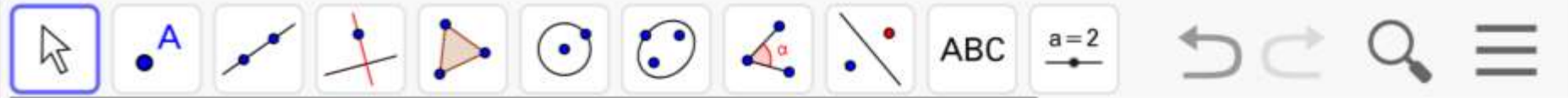
Consider the curve:  $r = 1.5 \cos \theta$

At  $\theta = 0, r = 1.5$

As  $\theta$  gets larger or smaller, so does  $r$ , but **not** in a linear fashion...

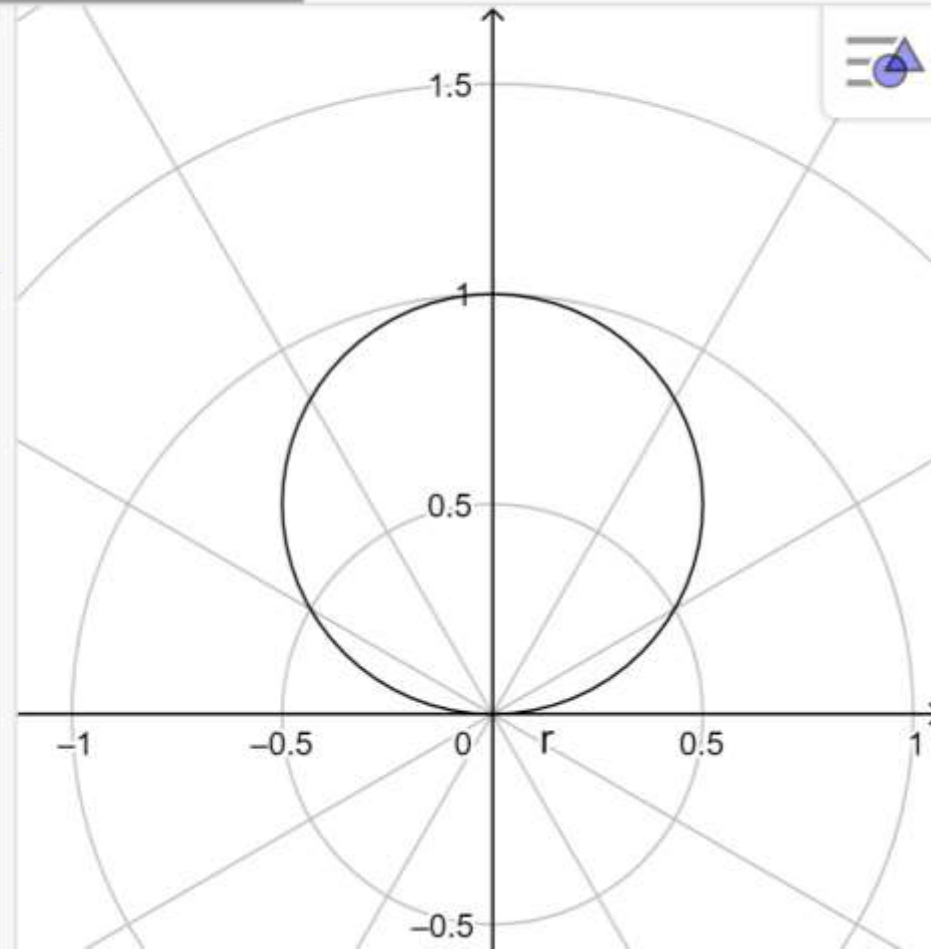






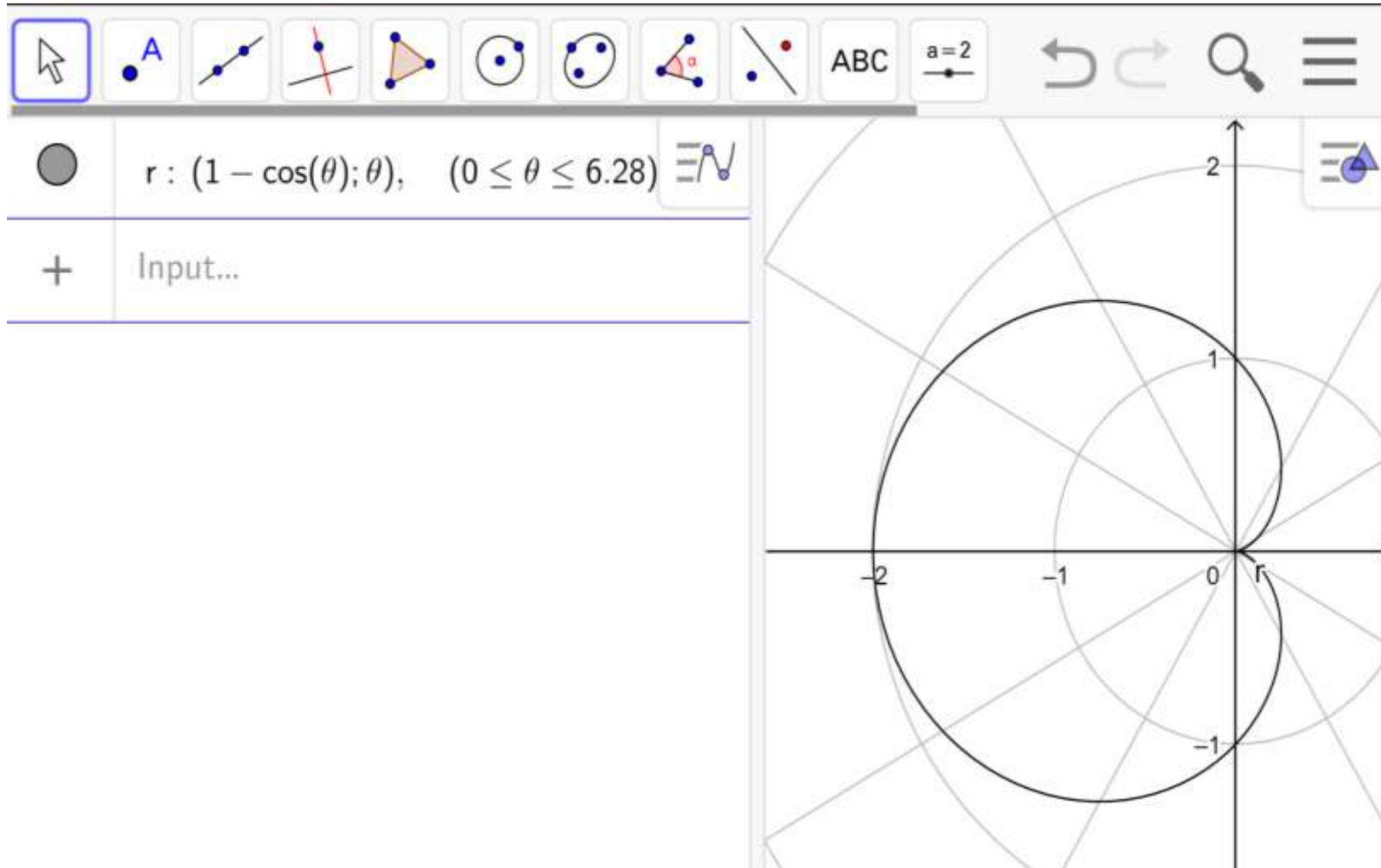
●  $r : (\sin(\theta); \theta), \quad (0 \leq \theta \leq 6.28)$

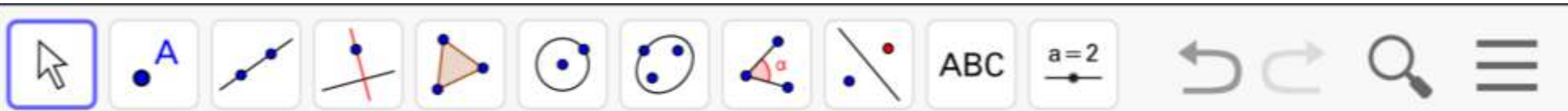
+ Input...



# Cardioid

Important in electrical  
engineering  
(microphones & digital  
sound processing)  
and mechanical  
engineering (rollers)

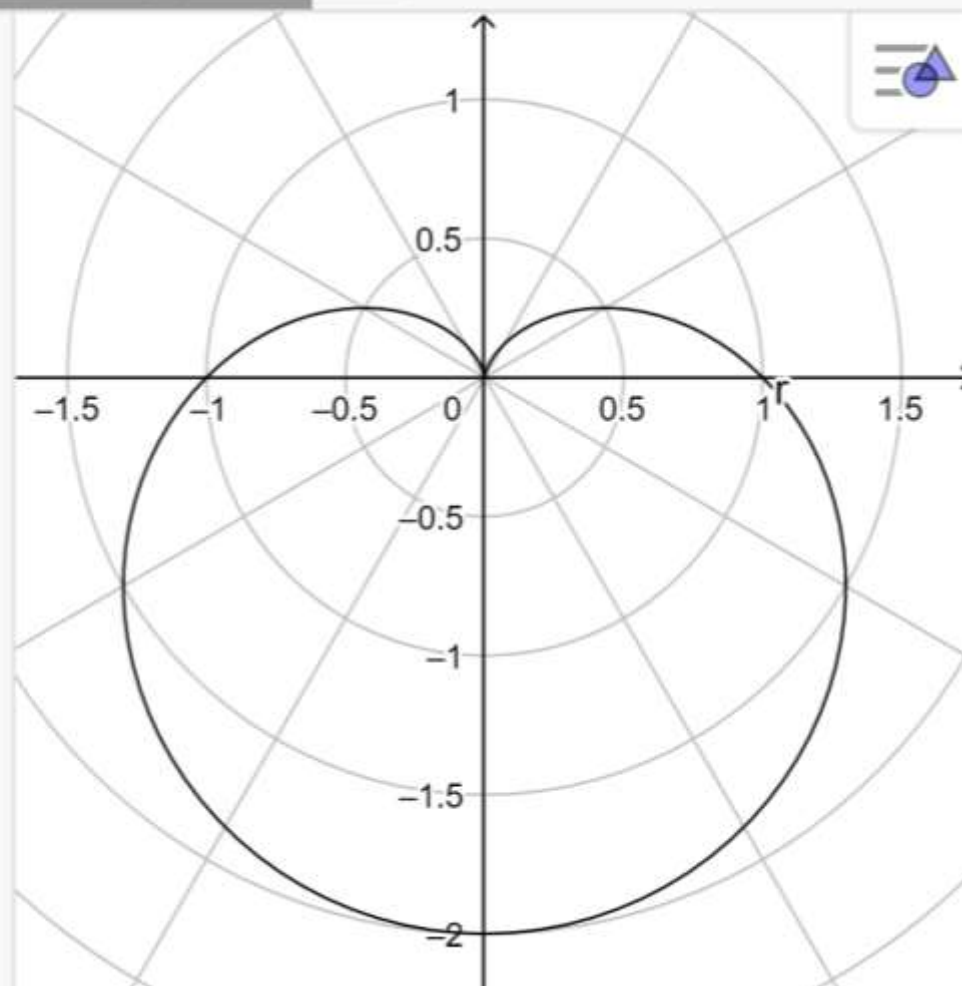


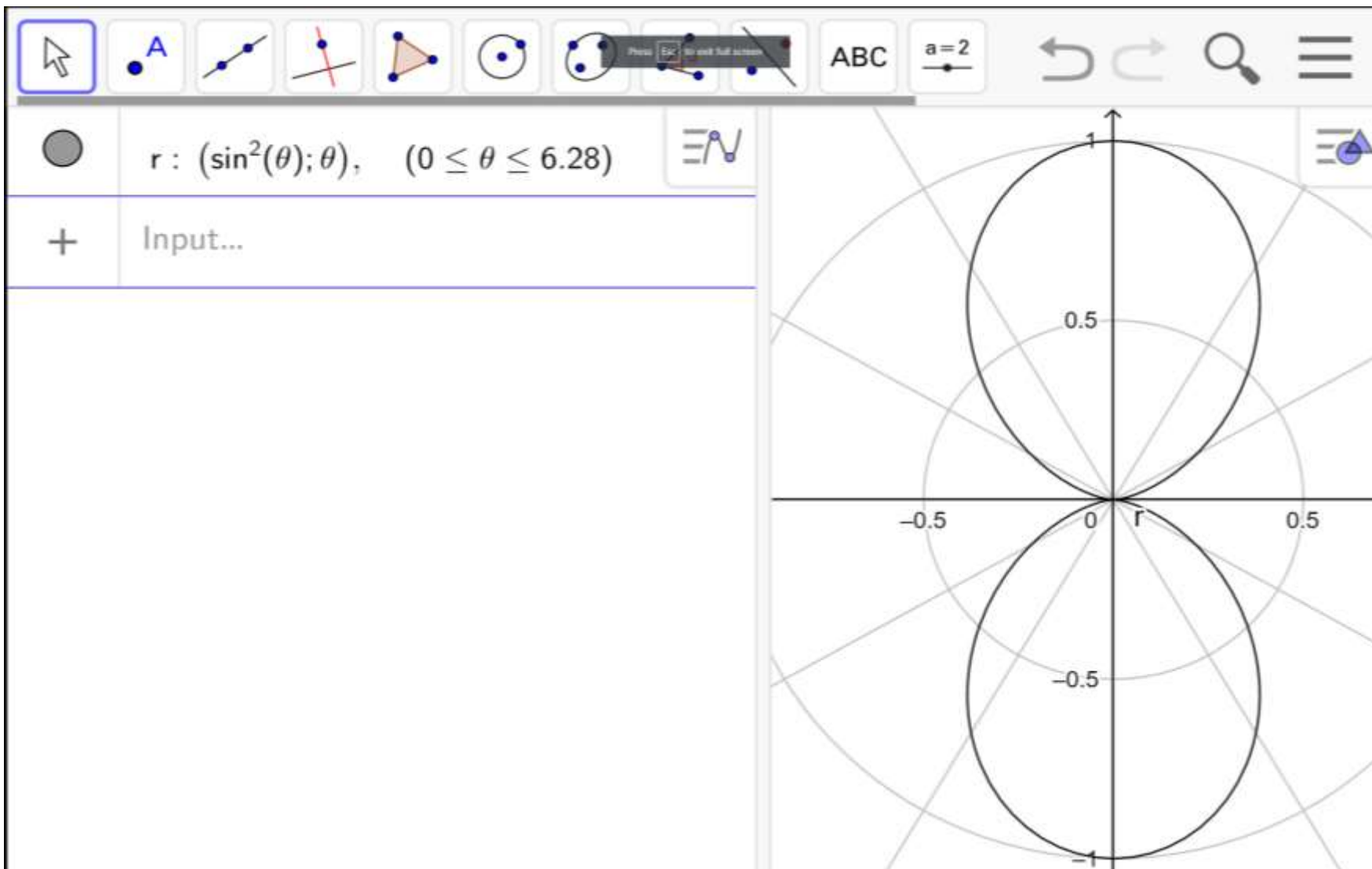


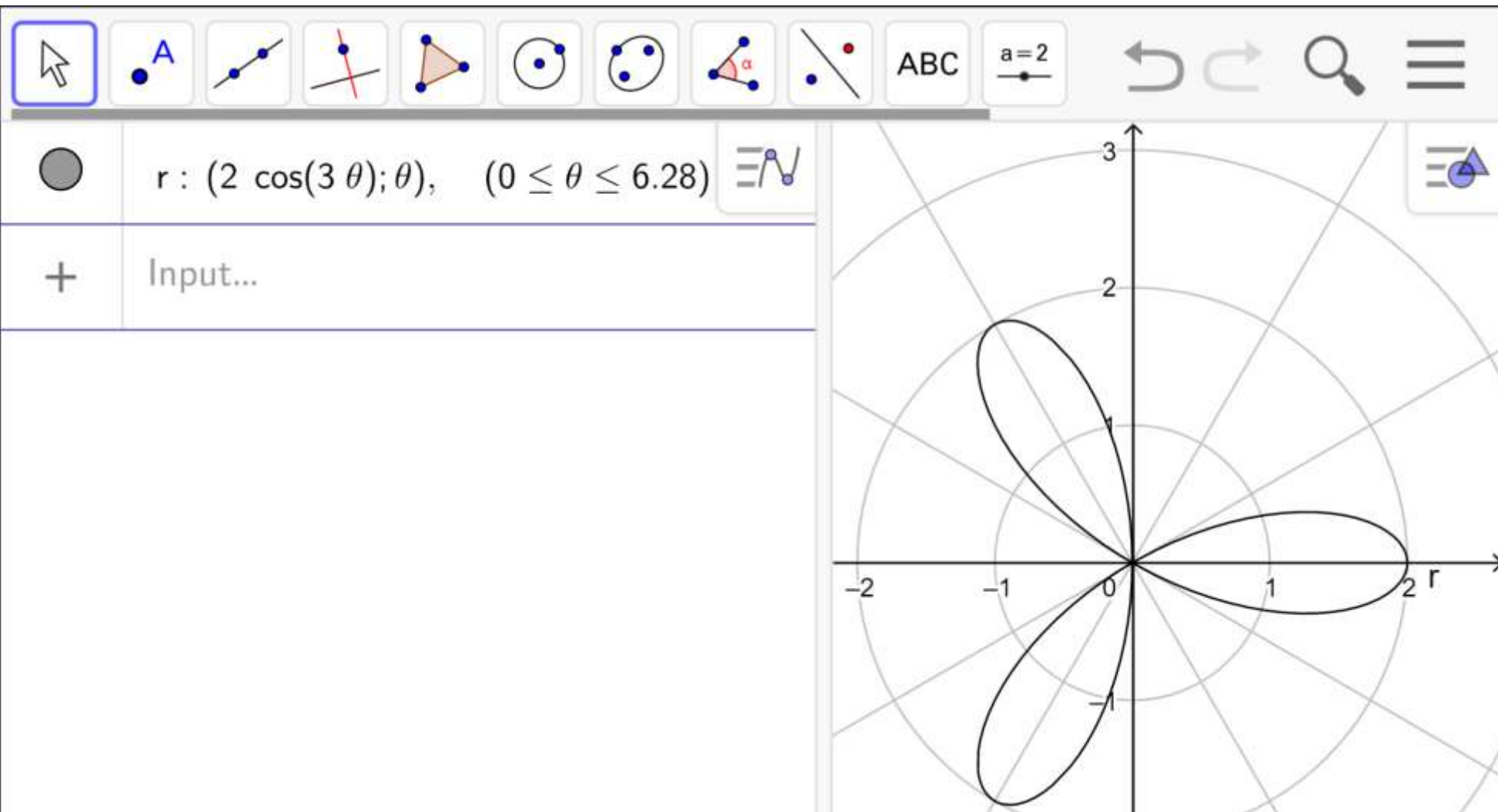
$$r : (1 - \sin(\theta); \theta), \quad (0 \leq \theta \leq 6.28)$$



Input...

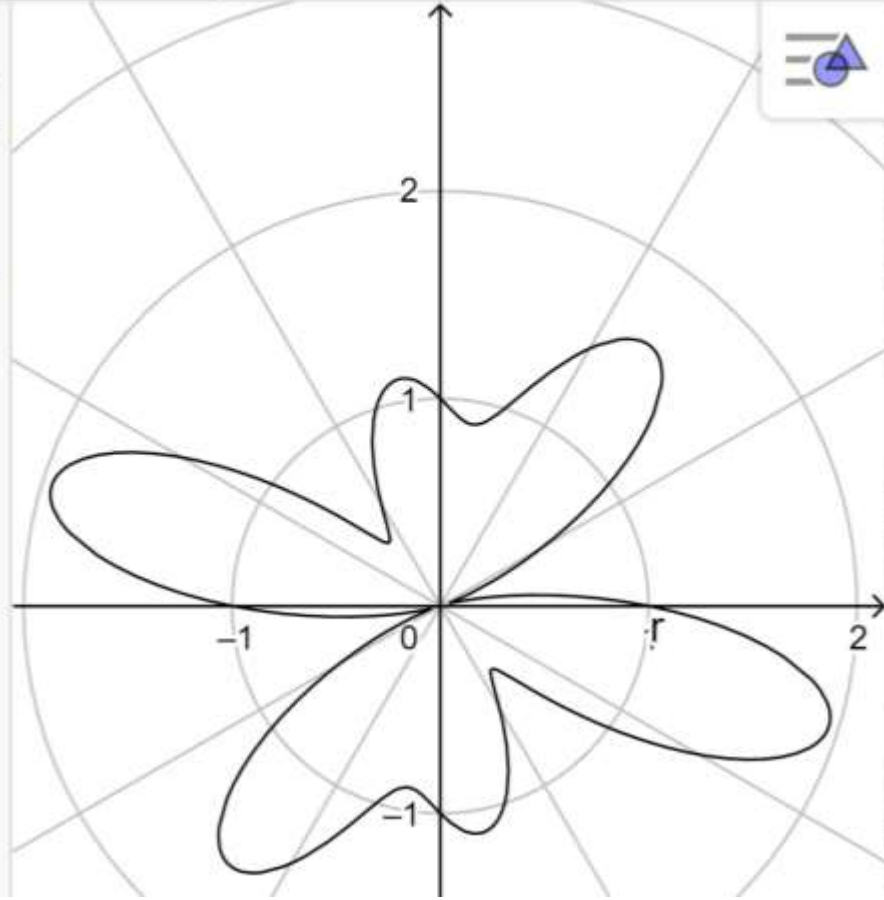






$$r : (1 - \cos(\theta) \sin(5 \theta); \theta), \quad (0 \leq \theta \leq 6.28318)$$

Input...



Some more help in Chapter 23, p279 but we go a bit beyond the book on this topic